

AT
FH
LB
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NB
YR
JG

Name: _____

Class: 12MT2__ or 12MTX__

Teacher: _____

CHERRYBROOK TECHNOLOGY HIGH SCHOOL



2007 AP4

YEAR 12 TRIAL HSC EXAMINATION

MATHEMATICS

*Time allowed - 3 HOURS
(Plus 5 minutes reading time)*

DIRECTIONS TO CANDIDATES:

- Attempt all questions.
- All questions are of equal value.
- Each question is to be commenced on a new page clearly marked Question 1, Question 2, etc on the top of the page. **.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used. Standard Integral Tables are provided
- Your solutions will be collected in one bundle stapled in the top left corner.

Please arrange them in order, Q1 to 10.

*****Each page must show your name and your class. *****

QUESTION 1**MARKS**

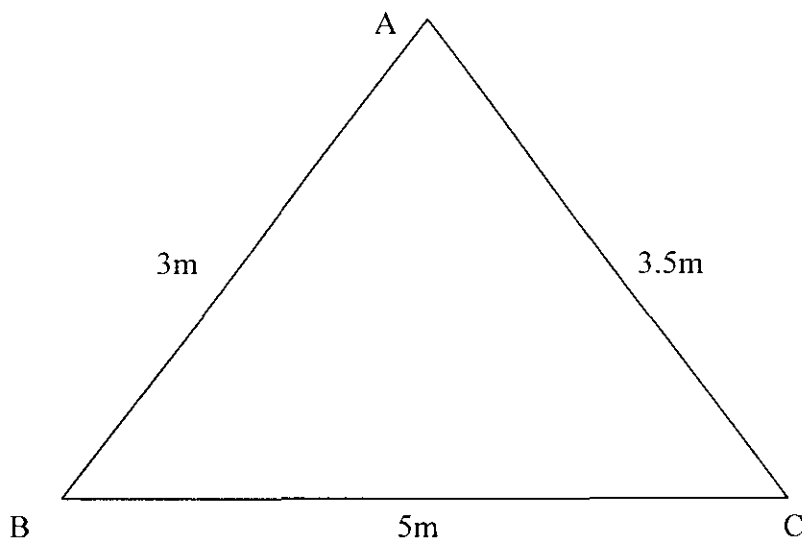
- (a) Evaluate to 2 decimal places $\frac{2 \cdot 34 \times 4 \cdot 654}{2 \cdot 786 - 1 \cdot 658}$ **2**
- (b) Express $\sqrt{27} + \sqrt{48}$ in the form $a\sqrt{3}$ where a is an integer **2**
- (c) Solve for x , $|x-4| < 6$ **2**
- (d) Find the exact value of $\tan \frac{2\pi}{3}$ **2**
- (e) Factorise $1+8x^3$ **2**
- (f) Solve $2x - \frac{3x+4}{5} = 9$ **2**

QUESTION 2 (Start a new page)**MARKS**(a) Differentiate with respect to x

(i) $\frac{x}{\ln x}$ 2

(ii) $(1 + \tan x)^5$ 2

(b)

NOT TO
SCALE

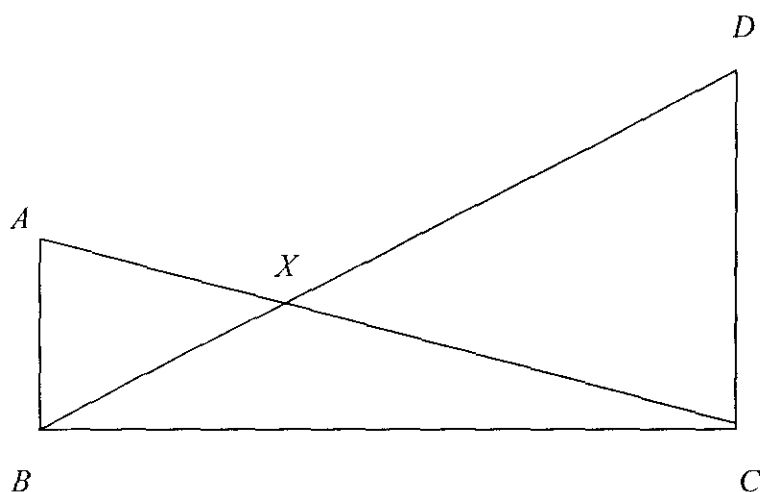
In the diagram above $AB = 3\text{m}$, $BC = 5\text{m}$ and $AC = 3.5\text{m}$. Find the size of the largest angle, correct to the nearest minute.

3

(c) (i) Find in exact form $\int_0^2 \frac{6x^2}{1+x^3} dx$ 3

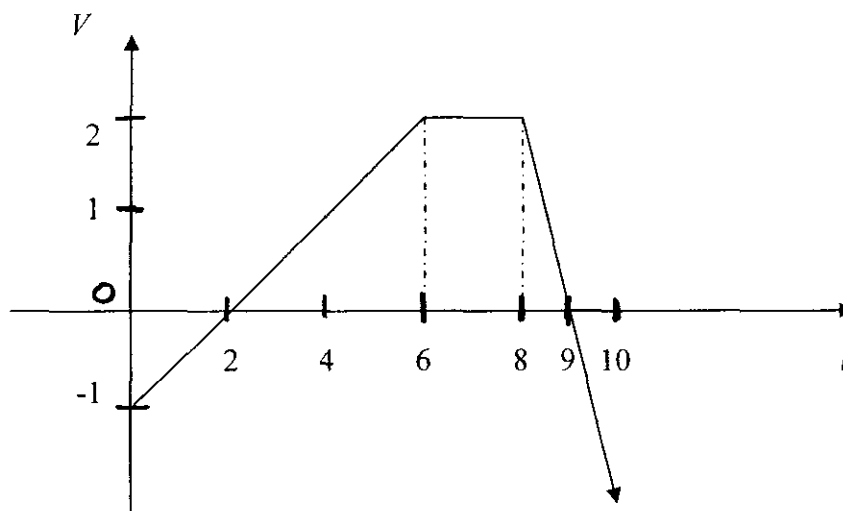
(ii) Find $\int \left(\frac{1}{3x^2} + e^{3x} \right) dx$ 2

- (a) The co-ordinates of points A , B and C are $A(1,4)$, $B(-2,1)$ and $C(2,-2)$.
- (i) Plot these points on a number plane 1
 - (ii) Show that the equation of the line AB is $x - y + 3 = 0$ 2
 - (iii) Find the perpendicular distance from C to AB 2
 - (iv) Find the distance AB (leave answer in simplest surd form) 1
 - (v) Hence, find the area of $\triangle ABC$, correct to one decimal place. 1
- (b) Show that $\log 2$, $\log 4$, $\log 8$, $\log 16$ form an arithmetic sequence 1
- (c) In the diagram AB is parallel to DC and $AB : DC = 2 : 3$



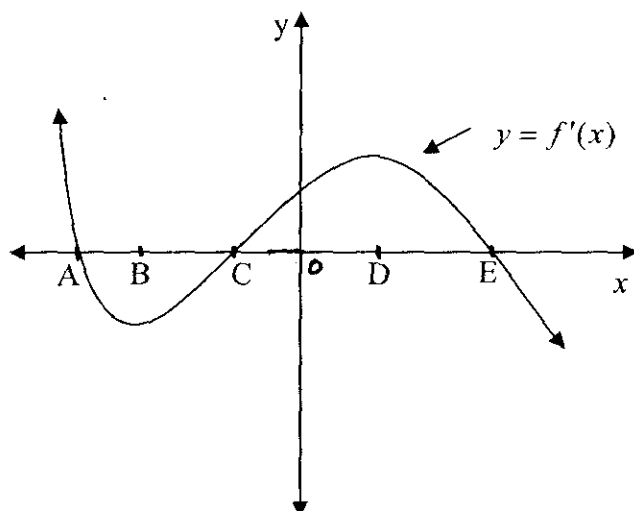
- (i) Show that $\triangle ABX \sim \triangle CDX$ 2
- (ii) Show that $9BX^2 = 4XD^2$ 2

- (a) A particle moves in a straight line with a velocity of $V \text{ ms}^{-1}$.
The graph of $V = f(t)$ is illustrated on the diagram below.



- | | | |
|---|---|---|
| (i) | When is the particle at rest? | 1 |
| (ii) | When does the particle first return to its starting point? | 1 |
| (iii) | Find the total distance travelled by the particle in the first 9 seconds. | 1 |
|
(b) The geometric series $1 - x + x^2 - \dots$ has a limiting sum of 4.
Find the value of x . | | 2 |
|
(c) Find the co-ordinates of the focus of the parabola $6y = x^2 - 2x - 11$ | | 3 |
|
(d) The quadratic equation $2x^2 - 3x - 6 = 0$ has roots α and β .
Without solving the quadratic equation, find the value of: | | |
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (iii) | $\alpha^2 + \beta^2$ | 2 |

(a)



The above diagram shows a sketch of the function $y = f'(x)$.
Sketch the function $y = f(x)$, indicating any stationary points.

2

(b) A function $f(x)$ is defined by $f(x) = x^3 - 3x^2 - 9x$.

(i) Find the turning points for the curve $y = f(x)$ and determine their nature.

3

(ii) Find any point(s) of inflexion.

2

(iii) Sketch the graph of $y = f(x)$ showing the turning points and point of inflexion.

1

(iv) Find the values of x for which both $f'(x) < 0$ and $f''(x) > 0$.

2

(c) Solve $2\sin^2 \theta - 1 = 0$ for $0 \leq \theta \leq 2\pi$

2

(a) If $y = A \cos 4x + B \sin 4x$, show that $\frac{d^2 y}{dx^2} + 16y = 0$

2

(b) The rate in litres per day at which a faulty tank is losing water is given by

$$R = 20 - \frac{20}{t+1}.$$

- (i) At what rate is the tank losing water after 4 days? 1
- (ii) What value does R approach as t becomes very large? 1
- (iii) Calculate the amount of water lost from the tank, in litres, over the first 9 days. (Leave your answer in exact form.) 2

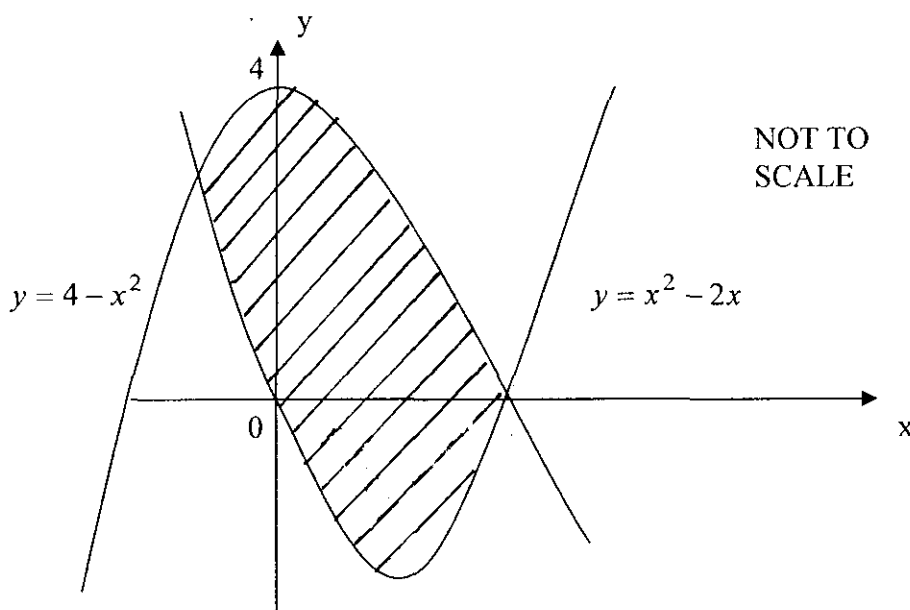
(c) The population P of a certain bacteria is falling according to the formula:

$$P = 3000e^{-kt}, \text{ where } t \text{ is in days.}$$

- (i) Show that $\frac{dP}{dt} = -kP$ 1
- (ii) If it takes 4 days for the number of bacteria to fall to 2000, what is the value of k (correct to 4 decimal places)? 2
- (iii) Find the rate of change of the population after one week. (Answer correct to 1 decimal place) 1
- (iv) How long will it take for the number of bacteria to fall to 10% of the initial number? (Answer correct to 1 decimal place) 2

- (a) Use Simpson's Rule with five function values to find an approximation for the value of $\int_0^1 10^x dx$. Give your answer correct to three decimal places. 2
- (b) (i) Show that $x = \frac{\pi}{8}$ is a solution of $\sin 2x = \cos 2x$ 1
- (ii) On the same set of axes, sketch the graphs of the functions $y = \sin 2x$ and $y = \cos 2x$ for $-\pi \leq x \leq \pi$. 3
- (iii) Hence, find graphically the number of solutions of $\tan 2x = 1$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. 1
- (iv) Use part (ii) to solve $\tan 2x \leq 1$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$. 1
- (c) The region enclosed between the curve $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, the x axis, and the lines $x = 4$ and $x = 11$ is rotated about the x axis. Calculate the exact volume of the solid generated. 4

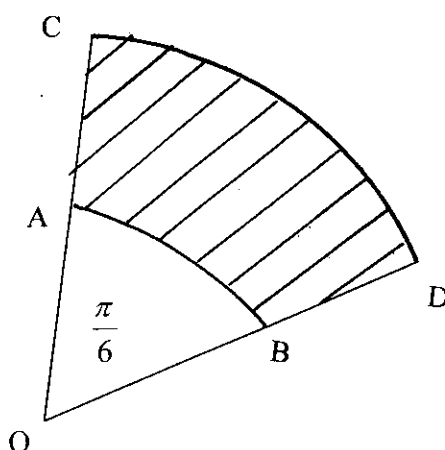
- (a) The graphs of the functions $y = 4 - x^2$ and $y = x^2 - 2x$.



- | | | |
|-------|---|---|
| (i) | Describe, using inequalities, the shaded region. | 2 |
| (ii) | By solving simultaneously, show that the points of intersection are at $x = -1$ and $x = 2$. | 1 |
| (iii) | Calculate the area of the shaded region. | 2 |
- (b) A particle moves along a straight line so that its distance x , in metres from a fixed point O is given by $x = t + \cos t$, where t is in the time measured in seconds.
- | | | |
|-------|---|---|
| (i) | Where is the particle initially? | 1 |
| (ii) | When, and where, does the particle first come to rest? | 3 |
| (iii) | When does the particle next come to rest? | 1 |
| (iv) | Find the acceleration of the particle when $t = \frac{\pi}{6}$ seconds. | 2 |

- (a) Simplify $\frac{\operatorname{cosec} A \sec A}{\tan A}$, expressing your answer in its simplest possible form. 1
- (b) Farmer Hay has hired a driller to drill a borehole to enable her to have access to the underground water on her property. The driller quotes a price of \$260 for the first 3 metres drilled, \$280 for the next 2 metres, \$300 for the next 2 metres and so on. The price increases by the same amount for each successive 2 metres of borehole drilled.
- (i) Show that the cost of drilling the portion from a depth of 25 metres to 27 metres is \$500. 2
- (ii) Calculate the total cost of drilling to a depth of 27 metres. 1
- (iii) The cost of drilling the borehole to reach water was \$12 500. 2
- Find the total depth drilled to give access to the water.
- (c) Bill and Ben decide to build a yacht to sail around the world. They approach a bank to see how much they can borrow. The bank manager tells them they can borrow \$P. They will receive an introductory rate of 6%p.a. for the first 3 months. The loan needs to be initially repaid in equal monthly repayments of \$3500 over 5 years and interest is charged monthly before each repayment.
- Let A_n be the amount owing by Bill and Ben at the end of the n th repayment.
- (i) Find an expression for A_1 1
- (ii) Show that $A_3 = P(1.005)^3 - 3500(1 + 1.005 + 1.005^2)$ 2
- (iii) At the end of the first three months Bill and Ben decide to fix the interest rate at 9% p.a. for the remainder of the loan. They make monthly repayments of \$4200 for the next 4.75 years. 3
- Calculate the value of \$P.

(a)

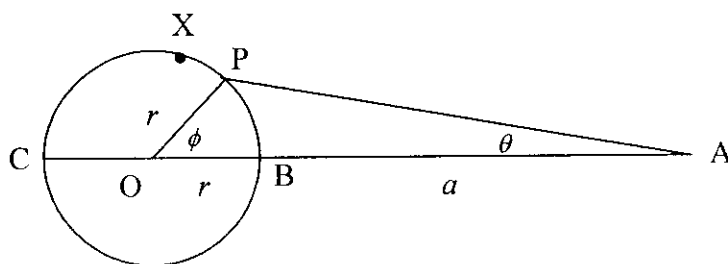


AB and CD are arcs of concentric circles with centre O.

2

$\angle AOB = \frac{\pi}{6}$ radians. $OA = 2$ centimetres. The shaded section has area $5\pi \text{ cm}^2$. Calculate the length of AC.

- (b) In the diagram below, the fixed points A, B, O, C, lie on a straight line. The points B, P, X, C are on the circumference of a circular pond, centre at O, radius r . AX is a tangent to the circle at X, and P lies on the arc BX.



The distance AB is a , $\theta = \angle PAB$ and $\phi = \angle POB$. The interval AP lies outside the pond. A boy, initially at A, is to walk in a straight line to P, at a constant speed $v_1 > 0$. He is then to run at a constant speed v_2 , where $v_2 > v_1$, along the shorter of the two arcs from P to C.

Question 10 (b) continued on the next page.....

- (i) Show that the total time
- T
- for the journey is

3

$$T = \frac{\sqrt{r^2 + (r+a)^2 - 2r(r+a)\cos\phi}}{v_1} + \frac{r(\pi - \phi)}{v_2}$$

- (ii) Find
- $\frac{dT}{d\phi}$

1

- (iii) By using parts (i), (ii) and the sine rule show that

2

$$\frac{dT}{d\phi} = \frac{r+a}{v_1} \left[\sin\theta - \frac{rv_1}{(r+a)v_2} \right]$$

- (iv) Given that
- $\angle AXO = 90^\circ$
- show that
- $\frac{dT}{d\phi} < 0$
- if
- P
- is at
- B
- ,

2

and $\frac{dT}{d\phi} > 0$ if P is at X .

- (v) Find, in terms of
- $\sin\theta$
- , the position of
- P
- which minimizes the time
- T
- .

2

END OF EXAM

Trial Paper 2007: Mathematics: Solutions

Q1. a) 9.654574468 ①
 9.65 ①

b) $3\sqrt{3} + 4\sqrt{3}$ ①
 $= 7\sqrt{3}$ ①

No marks for CFM

c) $-6 < x - 4 < 6$ ①
 $-2 < x < 10$ ①

d) $\tan \frac{2\pi}{3} = -\tan \frac{\pi}{3}$ ① for second quadrant (negative)
 $= -\sqrt{3}$ ① (aw 1 for $\sqrt{3}$)

e) $\underbrace{(1+2x)}_{①} \underbrace{(1-2x+4x^2)}_{①}$

f) $10x - 3x - 4 = 45$ ①
 $7x = 49$
 $x = 7$ ①

[aw 1 for CFM
ie $10x - 3x + 4 = 45$
 $x = \frac{41}{7} = 5.86 = 5\frac{6}{7}$]

Q2. a) (i) $\frac{d}{dx} = \frac{\ln x \times 1 - x \times \frac{1}{x}}{(\ln x)^2}$ ①
 $= \frac{\ln x - 1}{(\ln x)^2}$ ①

$\left[\frac{\ln x^2 \neq (\ln x)^2}{\ln^2 x} \right]$ - 1 mark

(ii) $\frac{d}{dx} = \underbrace{5(1+\tan x)^4}_{①} \times \underbrace{\sec^2 x}_{①} = 5\sec^2 x (1+\tan x)^4$

b) $\cos A = \frac{3^2 + 3 \cdot 5^2 - 5^2}{2 \times 3 \times 3 \cdot 5}$
 $(100^\circ 17' 11.6'')$
 $A = 100^\circ 17'$

① Incorrect formula = 0 marks
① incorrect sign
① $79^\circ 43'$ (-1 mark)

* cos rule after using sine rule
 $\Rightarrow$ ambiguous case $\Rightarrow$ 0 marks

(2)

Q2 cont'd

$$(c) \quad (i) \quad \int_0^2 \frac{6x^2}{1+x^3} dx = 2 \int_0^2 \frac{3x^2}{1+x^3} dx$$

$$= 2 \left[\ln(1+x^3) \right]_0^2 \quad (1)$$

$$= 2 \ln(1+2^3) - 2 \ln(1+0^3) \quad (1)$$

$$= 2 \ln 9 - 2 \ln 1$$

$$= 2 \ln 9 \quad (1) \quad \text{accept either}$$

$$= \ln 81$$

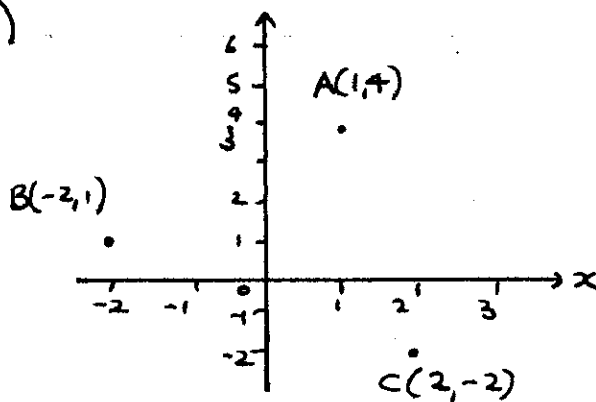
$$(ii) \quad \int \frac{1}{3x^2} + e^{3x} dx = \int \frac{1}{3} x^{-2} + e^{3x} dx$$

$$= \underbrace{-\frac{1}{3} x^{-1}}_{(1)} + \underbrace{\frac{e^{3x}}{3}}_{(1)} + C$$

$$= -\frac{1}{3x} + \frac{e^{3x}}{3} + C \quad \text{ISE}$$

made many mistake
 $\int \frac{1}{3x^2} = \log_e 3x^2$

Q3. a) i)



(1) all 3 points must be clearly labelled.

$$ii) \quad m_{AB} = \frac{4-1}{1+2} = \frac{3}{3} = 1 \quad (1)$$

$$\text{Eqn AB} \quad \left. \begin{aligned} y-4 &= 1(x-1) \\ y-4 &= x-1 \end{aligned} \right\} (1)$$

$$\therefore x - y + 3 = 0$$

$$iii) \quad d_{\perp} = \frac{|1 \times 2 - 1 \times -2 + 3|}{\sqrt{1^2 + (-1)^2}} \quad (1)$$

$$= \frac{7}{\sqrt{2}} = \frac{7\sqrt{2}}{2} \text{ units} \quad (1)$$

3

Q3 continued

(iv)

$$d_{AB} = \sqrt{(1+2)^2 + (4-1)^2}$$

$$= \sqrt{18}$$

$$= 3\sqrt{2} \text{ units} \quad (1)$$

$$(v) \quad A = \frac{1}{2} \times 3\sqrt{2} \times \frac{7}{\sqrt{2}}$$

$$= \frac{21}{2}$$

$$= 10.5 \text{ units}^2 \quad (1)$$

(b)

AP

$$T_2 - T_1 = T_3 - T_2$$

$$= T_2 - T_1$$

$$= \log 4 - \log 2$$

$$= 2\log 2 - \log 2$$

$$= \log 2$$

$$\text{RHS} = T_3 - T_2$$

$$= \log 8 - \log 4$$

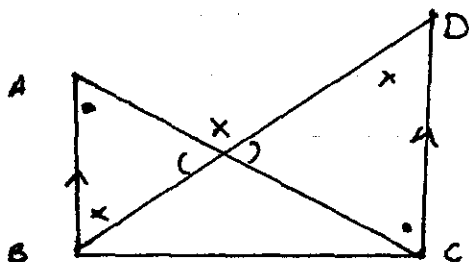
$$= 3\log 2 - 2\log 2$$

$$= \log 2$$

$$\therefore \text{LHS} = \text{RHS} \quad (1)$$

$\therefore \log 2, \log 4, \log 8, \log 16$ is an arithmetic sequence

(c)



$$AB : DC = 2 : 3$$

$$\therefore \frac{AB}{DC} = \frac{2}{3}$$

$$(i) \quad \left. \begin{array}{l} \hat{BAX} = \hat{XCD} \quad (\text{alternate } \angle's =, AB \parallel DC) \\ \hat{ABX} = \hat{XDC} \quad (\text{alternate } \angle's =, AB \parallel DC) \\ \hat{AXB} = \hat{DXC} \quad (\text{vertically opposite } \angle's =) \end{array} \right] \quad (1)$$

for any
2 correct
with reasons

$$\therefore \triangle AXB \parallel \triangle CXD \quad (\text{equiangular}) \quad (1)$$

$$(ii) \quad BX : XD = 2 : 3 \quad (\text{corresponding sides in similar } \triangle's) \quad (1)$$

$$\left. \begin{array}{l} \frac{BX}{XD} = \frac{2}{3} \\ 3BX = 2XD \end{array} \right] \quad (1)$$

$$\therefore 9BX^2 = 4XD^2$$

④

Q4 a) i) particle at rest when $v=0$
 $\therefore t=2, t=9$ seconds

①

ii) using areas: area below = area above
occurs when $t=4$ seconds

①

iii) Total distance = Sum of areas
 $= \frac{1}{2} \times 2 \times 1 + \frac{1}{2} \times 2 (2+7)$
 $= 10\text{m}$

①

b) $a=1$ $r=-x$ $S_{\infty}=4$ $S_{\infty}=\frac{a}{1-r}$
 $4=\frac{1}{1+x}$ ①

$$4x+4=1$$

$$4x=-3$$

$$x=-\frac{3}{4}$$

①

c) $6y+11+1=x^2-2x+1$

$$6(y+2)=(x-1)^2$$

concave up.

$\therefore$ vertex at $(1, -2)$

①

$$4a=6$$

$$a=\frac{3}{2} \text{ focal length } ①$$

$$\therefore \text{ focus } (1, -\frac{1}{2}) \quad ①$$

d) i) $\alpha+\beta=-\frac{b}{a}$ $\alpha+\beta=\frac{3}{2}$

ii) $\alpha\beta=\frac{c}{a}$ $\alpha\beta=-\frac{6}{2}=-3$

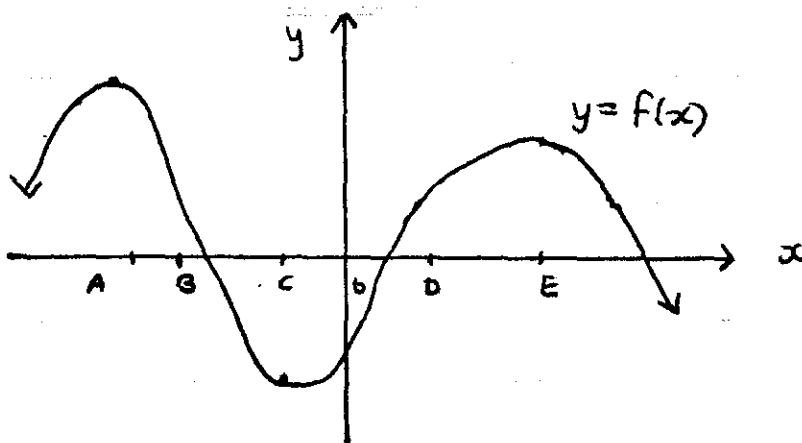
iii) $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta$ $\alpha^2+\beta^2=(\frac{3}{2})^2-2(-3)$ ①

$$=8\frac{1}{4}$$

①

⑤

Q5. a)



- ① shape
- ① stationary points at A, C, E

b)

$$f(x) = x^3 - 3x^2 - 9x$$

$$f'(x) = 3x^2 - 6x - 9$$

$$f''(x) = 6x - 6$$

i) $f'(x) = 0$

$$3(x^2 - 2x - 3) = 0$$

$$(x - 3)(x + 1) = 0$$

$$\therefore x = 3, -1$$

- ① * award full marks for correct x values and correct tests (ignore y-value)

When $x = 3$ $f(3) = 27 - 27 - 27 = -27$

test $x = 3$ $f''(x) = 18 - 6 = 12 > 0$ $\therefore$ minimum turning point at $(3, -27)$

When $x = -1$ $f(-1) = -1 - 3 + 9 = 5$

test $x = -1$ $f''(x) = -6 - 6 = -12 < 0$ $\therefore$ maximum turning point at $(-1, 5)$

ii) $f''(x) = 0$

$$6x - 6 = 0$$

$$x = 1$$

When $x = 1$ $f(1) = 1 - 3 - 9 = -11$

- ① test
- ① point

test $x = 1$

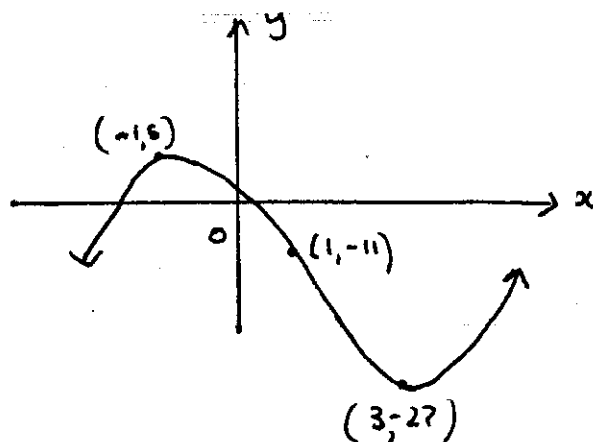
x	0	1	2
f''(x)	-6	0	6

$\therefore$ change in concavity

$\therefore$ point of inflexion occurs at $(1, -11)$

(6)

Q5 continued. iii)



① for correct shape and labelled points (need to include y values)

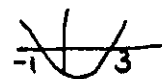
$$(iv) \quad f'(x) < 0 \quad x^2 - 2x - 3 < 0$$

$$-1 < x < 3$$

$$f''(x) > 0 \quad 6x - 6 > 0$$

$$x > 1$$

①



∴ for both to hold:

$$1 < x < 3 \quad ①$$

$$c) \quad 2 \sin^2 \theta - 1 = 0$$

$$\sin \theta = \pm \frac{1}{\sqrt{2}} \quad ①$$

$$\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$$

ie $\frac{1}{\sqrt{2}}$ only

①

AW ① for answers in degrees

cfm → aw 1 mark for 2 answers (must have all 4 solutions)

Q6. a) $y = A \cos 4x + B \sin 4x$

$$\frac{dy}{dx} = -4A \sin 4x + 4B \cos 4x$$

$$\frac{d^2y}{dx^2} = -16A \cos 4x - 16B \sin 4x$$

①

$$LHS = \frac{d^2y}{dx^2} + 16y$$

$$= -16A \cos 4x - 16B \sin 4x + 16(A \cos 4x + B \sin 4x)$$

$$= -16A \cos 4x - 16B \sin 4x + 16A \cos 4x + 16B \sin 4x \quad ①$$

$$= 0$$

$$= RHS$$

⑦

Q 6 continued

b) i) $R = 20 - \frac{20}{t+1}$

$t = 4 \quad R = 20 - \frac{20}{4+1}$

$R = 16 \text{ L/day}$

①

ii) as $t \rightarrow \infty \quad -\frac{20}{t+1} \rightarrow 0$

$\therefore R \rightarrow 20$

$\therefore$ rate approaches 20 L/day ①

iii) $\int_0^9 20 - \frac{20}{t+1} dt = \left[20t - 20 \ln(t+1) \right]_0^9$ ①

$= (180 - 20 \ln 10) - (0 - 20 \ln 1)$

$= 180 - 20 \ln 10$

needed exact

①

c) i) $P = 3000 e^{-kt}$

$\frac{dP}{dt} = -k 3000 e^{-kt}$

$= -kP$

①

ii) $2000 = 3000 e^{-4k}$

①

$\frac{2}{3} = e^{-4k}$

$\ln \frac{2}{3} = -4k$

$k = -\frac{1}{4} \ln \frac{2}{3}$

$= 0.101366277$

$= 0.1014$

①

iii) $\frac{dP}{dt} = -(-\frac{1}{4} \ln \frac{2}{3}) \times 3000 e^{-(-\frac{1}{4} \ln \frac{2}{3}) \times 7}$

$= -149.5736347$

$= -149.6$

①

OR

using $k = 0.1014$

$= -149.5880795$

$= -149.6$

iv) 10% of 3000 = 300

*using $k = 0.1014$

$t = 22.70793977$

① $\left[\begin{array}{l} 300 = 3000 e^{-kt} \\ \frac{1}{10} = e^{-kt} \\ t = -\frac{1}{k} \ln \frac{1}{10} \end{array} \right]$

$t = 22.71549135$

$t = 22.7 \text{ days}$

using k in full

①

(8)

Q7. a) $y_0 \quad y_1 \quad y_2 \quad y_3 \quad y_4$
 $0 \quad 0.25 \quad 0.5 \quad 0.75 \quad 1.0$

$$\int_0^1 10^x dx = \frac{0.25}{3} [(1+10) + 4(10^{0.25} + 10^{0.75}) + 2(10^{0.5})] \quad (1)$$

$$= 3.910943831 \quad (1)$$

$$= 3.911$$

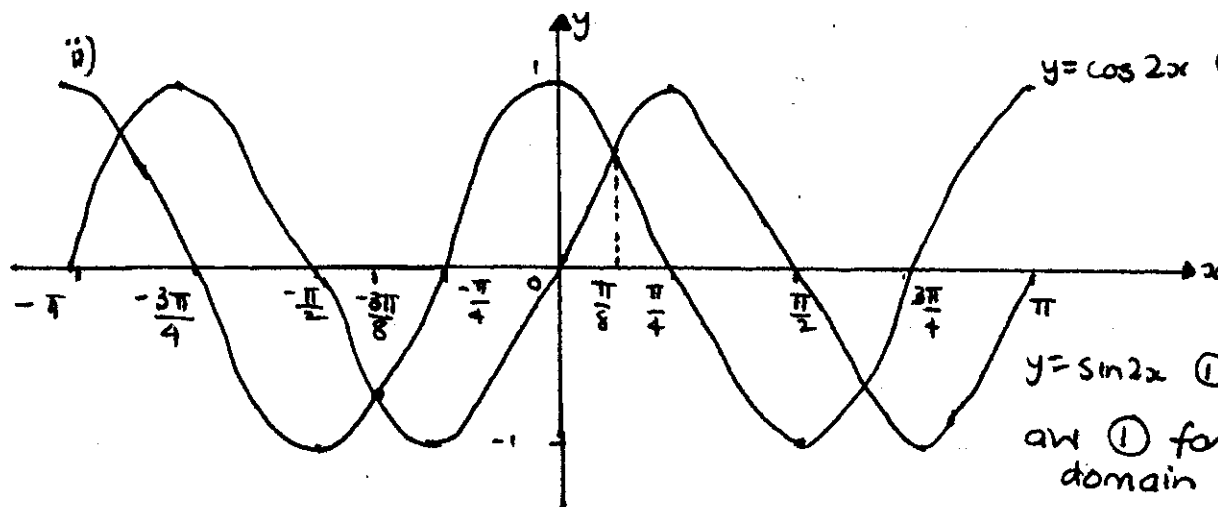
b) i) LHS = $\sin 2x$
 $= \sin \frac{2\pi}{8}$
 $= \frac{1}{\sqrt{2}}$

RHS = $\cos 2x$
 $= \cos \frac{2\pi}{8}$
 $= \frac{1}{\sqrt{2}}$

need both
for mark
(1)

$\therefore$ LHS = RHS

also $\frac{\sin 2x}{\cos 2x} = 1$
 $\tan 2x = 1$
 $2x = \frac{\pi}{4}$
 $x = \frac{\pi}{8}$



* maximum (1) for amplitude 2 or for $\sin x, \cos x$ graphs in the correct domain

(iii) 2 points of intersection

$\therefore \tan 2x = 1 \quad -\frac{\pi}{2} < x < \frac{\pi}{2}$

$\therefore$ 2 solutions

(1) accept $-\frac{3\pi}{8}, \frac{\pi}{8}$

(iv) $-\frac{3\pi}{8} \leq x \leq \frac{\pi}{8}$

(1)

(9)

Q7 continued

$$y = \sqrt{x} + \frac{1}{\sqrt{x}}$$

$$V = \pi \int_a^b y^2 dx$$

$$y^2 = (\sqrt{x} + \frac{1}{\sqrt{x}})(\sqrt{x} + \frac{1}{\sqrt{x}})$$

$$y^2 = x + 2 + \frac{1}{x} \quad (1)$$

$$\begin{aligned} \therefore \text{Volume} &= \pi \int_4^{11} x + 2 + \frac{1}{x} dx \\ &= \pi \left[\frac{x^2}{2} + 2x + \ln x \right]_4^{11} \quad (1) \\ &= \pi \left[\left(\frac{121}{2} + 22 + \ln 11 \right) - \left(\frac{16}{2} + 2 \times 4 + \ln 4 \right) \right] \quad (1) \\ &= \pi \left[\frac{133}{2} + \ln 11 - \ln 4 \right] \quad (1) \\ &= \pi \left[\frac{133}{2} + \ln \frac{11}{4} \right] \end{aligned}$$

Q8 a) (i) $y = 4 - x^2$
 test (0,0) $0 \leq 4 - 0^2$ True poorly attempted
 $\therefore y \leq 4 - x^2 \quad (1)$

$y = x^2 - 2x$
 Test (1,0) $0 \geq 1 - 2$
 $\therefore y \geq x^2 - 2x \quad (1)$

ii) $4 - x^2 = x^2 - 2x$
 $2x^2 - 2x - 4 = 0$
 $2(x^2 - x - 2) = 0$
 $(x - 2)(x + 1) = 0 \quad (1)$
 $\therefore x = 2, -1$ as required

iii) Area = $\int_{-1}^2 4 - x^2 - x^2 + 2x dx$
 $= \int_{-1}^2 4 + 2x - 2x^2 dx$
 $= \left[4x + x^2 - \frac{2x^3}{3} \right]_{-1}^2 \quad (1)$
 $= \left[\left(8 + 4 - \frac{16}{3} \right) - \left(-4 + 1 + \frac{2}{3} \right) \right] = 9 \text{ units}^2 \quad (1)$

(10)

Q8 continued

b) i) $t=0$ $x = 0 + \cos 0$
 $x = 1 \text{ m}$ (1) (i.e. 1m to right of origin)

ii) Particle at rest $v=0$ $x = t + \cos t$
 $\frac{dx}{dt} = 1 - \sin t$ (1)

$$1 - \sin t = 0$$

$$\sin t = 1$$

$$t = \frac{\pi}{2} \text{ seconds} \quad (1)$$

When $t = \frac{\pi}{2} \text{ sec}$ $x = \frac{\pi}{2} + \cos \frac{\pi}{2}$
 $= \frac{\pi}{2} \text{ metres} \quad (1)$
 $[= 1.570796327]$

iii) $t = \frac{5\pi}{2} \text{ seconds}$ (i.e. $\frac{\pi}{2} + 2\pi$) (1)

iv) $\frac{d^2x}{dt^2} = -\cos t$ (1)

When $t = \frac{\pi}{6}$ $a = -\cos \frac{\pi}{6}$
 $= -\frac{\sqrt{3}}{2} \text{ ms}^{-2} \quad (1)$ Accept -0.866025403
 -0.866

Q9. a) $\frac{\csc A \sec A}{\tan A} = \frac{1}{\sin A \cos A} \times \frac{\cos A}{\sin A}$
 $= \frac{1}{\sin^2 A}$
 $= \csc^2 A \quad (1)$

b) Depth 3, 5, 7,
 Cost 260, 280, 300,

i) Depth $a=3$ $d=2$ $T_n=27$
 $27 = 3 + (n-1) \times 2$
 $n-1=12 \quad \therefore n=13 \quad (1)$

(11)

Q 9 continued

Cost $a = 260$ $d = 20$ $n = 13$

$$T_{13} = 260 + (13-1) \times 20 \quad (1)$$

$$= \$500$$

$$(ii) \quad S_{13} = \frac{13}{2} (2 \times 260 + (13-1) \times 20) \quad (1)$$

$$= \$4940$$

$$(iii) \quad S_n = 12500 \quad 12500 = \frac{n}{2} (520 + (n-1) \times 20)$$

$$25000 = n(520 + 20n - 20)$$

$$25000 = 500n + 20n^2$$

$$n^2 + 25n - 1250 = 0$$

$$(n-25)(n+50) = 0$$

$$n = 25, -50 \quad \text{but } n > 0$$

$$\therefore n = 25. \quad (1)$$

Depth $T_{25} = 3 + (25-1) \times 2$

$$= 51m \quad (1)$$

$$(c) \quad i) \quad A_1 = P(1.005) - 3500 \quad (1)$$

$$ii) \quad A_2 = A_1(1.005) - 3500$$

$$= P(1.005)^2 - 3500(1 + 1.005) \quad (1)$$

$$A_3 = A_2(1.005) - 3500$$

$$= P(1.005)^3 - 3500(1.005)^2 - 3500(1.005) - 3500 \quad (1)$$

$$\therefore A_3 = P(1.005)^3 - 3500(1 + 1.005 + 1.005^2)$$

$$iii) \quad A_4 = A_3(1.0075) - 4200$$

$$A_5 = A_4(1.0075) - 4200$$

$$= (A_3(1.0075) - 4200)1.0075 - 4200$$

$$= A_3(1.0075)^2 - 4200(1.0075) - 4200$$

$$= A_3(1.0075)^2 - 4200(1.0075 + 1)$$

$$\therefore A_6 = A_3(1.0075)^3 - 4200(1 + 1.0075 + 1.0075^2)$$

P

$$r = 6\% \text{ p.a.} = 0.5\% \text{ p.m.}$$

$$n = 3 \text{ months}$$

$$\text{repayments} = 3500$$

$$r = 9\% \text{ p.a.} = 0.75\% \text{ p.m.}$$

$$n = 57 \text{ months}$$

$$\text{repayments} = 4200$$

12

Q9 continued

$$(c) \text{ iii) } A_{60} = A_3 (1.0075)^{57} - 4200 (1 + 1.0075 + 1.0075^2 + \dots + 1.0075^{56}) \quad \textcircled{1}$$

but loan is repaid $\therefore A_{60} = 0$

geometric series
a=1 r=1.0075 n=57

$$0 = A_3 (1.0075)^{57} - 4200 \times \frac{(1.0075^{57} - 1)}{1.0075 - 1}$$

$$A_3 (1.0075)^{57} = \frac{4200 (1.0075^{57} - 1)}{1.0075 - 1}$$

from (ii) use A_3

$$P \left[(1.005)^3 - 3500 (1 + 1.005 + 1.005^2) \right] (1.0075)^{57} = \frac{4200 (1.0075^{57} - 1)}{1.0075 - 1} \quad \textcircled{1}$$

$$\therefore P = \left[\frac{4200 (1.0075^{57} - 1)}{(1.0075 - 1) (1.0075)^{57}} + 3500 (1 + 1.005 + 1.005^2) \right] \div (1.005)^3$$

$$P = 201731.5044 \quad \textcircled{1} \quad (\text{1SE})$$

$$P = \$201731.50$$

$$P = \$201732$$

Q10. a) area of segment = $\frac{1}{2} \theta (R^2 - r^2)$

$$5\pi = \frac{1}{2} \times \frac{\pi}{6} (R^2 - 2^2)$$

$$60 = R^2 - 4$$

$$R^2 = 64$$

$$R = 8 \quad \text{as } R > 0 \quad \textcircled{1}$$

$$\therefore OC = 8 \text{ cm}$$

$$\therefore AC = 8 - 2 = 6 \text{ cm.} \quad \textcircled{1} \quad \text{mark only awarded for correct answer}$$

(13)

Q10 continued.

b) i) distance AP (cosine rule)

$$AP^2 = r^2 + (r+a)^2 - 2r(r+a) \cos \phi$$

$$AP = \sqrt{r^2 + (r+a)^2 - 2r(r+a) \cos \phi}$$

$$\text{time taken} = \frac{AP}{v_1} = \frac{\sqrt{r^2 + (r+a)^2 - 2r(r+a) \cos \phi}}{v_1} \quad (1)$$

distance PC (arc length)

$$l = r \theta = r \times (\pi - \phi) \quad (\text{COP})$$

$$\text{time taken} = \frac{PC}{v_2} = \frac{r(\pi - \phi)}{v_2} \quad (1)$$

$$\begin{aligned} \text{total time (T)} &= \frac{AP}{v_1} + \frac{PC}{v_2} \\ &= \left[\frac{\sqrt{r^2 + (r+a)^2 - 2r(r+a) \cos \phi}}{v_1} + \frac{r(\pi - \phi)}{v_2} \right] \quad (1) \end{aligned}$$

$$\begin{aligned} \text{ii)} \quad \frac{dT}{d\phi} &= \frac{1}{2} \left(\frac{r^2 + (r+a)^2 - 2r(r+a) \cos \phi}{v_1} \right)^{-\frac{1}{2}} \times 2r(r+a) \sin \phi - \frac{r}{v_2} \\ &= \frac{r(r+a) \sin \phi}{v_1 \sqrt{r^2 + (r+a)^2 - 2r(r+a) \cos \phi}} - \frac{r}{v_2} \quad (1) \end{aligned}$$

$$\text{iii)} \quad \text{using sine rule: } \frac{r}{\sin \theta} = \frac{AP}{\sin \phi} \Rightarrow \frac{\sin \phi}{AP} = \frac{\sin \theta}{r}$$

$$\frac{dT}{d\phi} = \frac{r(r+a) \sin \phi}{v_1 AP} - \frac{r}{v_2} \quad (\text{using (i)}) \quad (1)$$

$$= \frac{r(r+a)}{v_1} \times \frac{\sin \theta}{r} - \frac{r}{v_2} \quad (1)$$

$$= \frac{r+a}{v_1} \left[\sin \theta - \frac{rv_1}{v_2(r+a)} \right]$$

(14)

Q10 continued

(c) (iv) When P is at B $\theta = 0$

$$\begin{aligned}\frac{dT}{d\phi} &= \frac{r+a}{v_1} \left[\sin\theta - \frac{rv_1}{v_2(r+a)} \right] \\ &= -\frac{r}{v_2} < 0 \quad \text{since } v_2 > 0 \quad (1)\end{aligned}$$

When P is at X $\hat{A}XO = 90^\circ \therefore \sin\theta = \frac{r}{r+a}$

$$\begin{aligned}\frac{dT}{d\phi} &= \frac{r+a}{v_1} \left[\frac{r}{r+a} - \frac{rv_1}{v_2(r+a)} \right] \\ &= \frac{r}{v_1} - \frac{r}{v_2} \\ &= \frac{r(v_2 - v_1)}{v_1 v_2} > 0 \quad \text{since } v_2 > v_1 > 0 \quad (1)\end{aligned}$$

(v). minimum $\Rightarrow$ stationary point $\rightarrow \frac{dT}{d\phi} = 0$

$$0 = \frac{r+a}{v_1} \left[\sin\theta - \frac{rv_1}{v_2(r+a)} \right]$$

$$\therefore \sin\theta = \frac{rv_1}{v_2(r+a)}$$

using (iv) Test

$\sin\theta$	at B	at P	at X
$\frac{dT}{d\phi}$	< 0	$= 0$	> 0
	-	0	+
	min		

$\therefore$ minimum occurs when $\sin\theta = \frac{rv_1}{v_2(r+a)} \quad (1)$